

ZS-M48

Suzuki's Finite-Interval Operator on Dirichlet Channels, and Character Decomposition as Numerical Preconditioning

A Channel Perturbation Theorem, the Frobenius-Resolved Kernel, and a Certified V_4 Case Study for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$

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This is a compact rewrite. Versions v1.0–v1.9 were internal audit drafts; every correction made during that process is recorded in the Correction History Supplement and is not repeated in the body.

Verification: one self-contained fail-closed suite, `zs_m48_verify_v2_7.py` — **380 checks**; the expected count is **hard-coded** (no environment override) and the script exits non-zero on the first failure. **Theorem-tier** blocks decided in arb ball arithmetic: the enclosures of §4, the real-zero certification of §4.3, the certified zero brackets, the interval Cholesky of §5, the budget inversions of §6, and the Frobenius resolution of §8.1. **VERIFIED** (not arb-certified): the perturbation norms $\|v_{\chi,a}\|_{TV}$ of Table 8.1 (Theorem T11 itself is **PROVEN**; only the numerical values of its bound are floating-point), and the threshold-law residuals of Table 6.1. **DIAGNOSTIC**: the sampled screw-Gram and Schoenberg matrices, the exploratory null model, the detected-zero list, and the fixed-point count. `--fast` is a smoke test, **not** the public certificate. **Provenance**, stated exactly: the analytic blocks of this version were executed to completion in isolation; a single-process run of all 380 checks takes roughly 15 minutes and exceeded the interactive limit of the drafting environment, so its stdout, JSON ledger, library versions, script SHA-256 and wall-clock time are **not yet included**. They are required before public release, and the hard-coded guard is what prevents a partial run from being reported as a complete one.

§0. Abstract

Suzuki attached to a Dirichlet series in the extended Selberg class a real even kernel g_F — the **screw function** — whose non-positivity is equivalent to GRH for F , and built a finite-interval self-adjoint operator from it [1–3]. This paper takes that machinery to a **decorated** object: the Dedekind zeta of the biquadratic field $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$, whose genus characters have conductors 1, 3, 11, 33. **Five** results follow. The first is an operator theorem; the last is the reason to read the paper.

(0) The channel perturbation theorem — Suzuki's operator, transferred. Suzuki proves that the finite-interval quadratic form of the ζ screw kernel is closed, lower semibounded, and has a self-adjoint Friedrichs extension with discrete spectrum [3]. That theorem does **not** transfer to Dirichlet L-functions for free. We prove that it does, and cheaply. Writing $h_\chi := g_\chi - g_\zeta$, the two archimedean densities have the **same** $1/(2|x|)$ singularity at the origin, so it cancels in the difference: $\omega_{\{c_\chi\}}(x) - \omega_{\{1/4\}}(x) \rightarrow 1/4 - c_\chi$. Consequently, on any $[-2a, 2a]$,

$h_\chi'' = v_{\chi,a}$, a **FINITE signed Radon measure**: finitely many prime atoms ($n \leq e^{2a}$), a point mass of **exactly $-\log q_\chi$** at the origin, and a bounded density.

Since the form difference is $q_\chi[u] - q_\zeta[u] = -\langle v_{\chi,a}, u * \tilde{u} \rangle$ and $|(u * \tilde{u})(t)| \leq \|u\|^2$, the perturbation is

L^2 -bounded with explicit norm $\|v_{\chi,a}\|_{TV}$. By KLMN, q_χ is closed and lower semibounded on the **same form domain** as q_ζ ; the associated operator is $A_a^\chi = A_a^\zeta + B_{\chi,a}$ with $\|B\| \leq \|v\|_{TV}$, so it is self-adjoint,

has **compact resolvent and discrete spectrum**, and its eigenvalues satisfy $|\lambda_n(A_a^\chi) - \lambda_n(A_a^\zeta)| \leq \|\chi_a\|_{TV}$ for every n . **No new compactness proof is required for each channel** — the perturbation lemma itself is of course new operator analysis.

(i) The abelian screw decomposition. For any finite abelian $K/\mathbb{Q}$ with Galois group G , $g_K = \sum_{\chi \in \hat{G}} g_\chi$, and the prime-power coefficient of g_K reads the splitting type: $\sum_\chi \chi(\text{Frob}_{p^m}) = |G| \cdot 1[\text{Frob} = e]$ at unramified p . **The screw kernel of an abelian field is a Frobenius detector.** All constants are derived, not fitted: the linear coefficient is $\beta_\chi = (L'/L)(\frac{1}{2}, \chi) = -\frac{1}{2}(\psi(c_\chi) - \log(\pi/q_\chi))$, forced by $\Lambda'(\frac{1}{2}, \chi) = 0$.

(ii) The kink measure. g_χ is not analytic data but a distribution: $g_\chi'' = \mu_\Lambda^\chi + 2\beta_\chi \delta_0 + T\{c_\chi\} - (\text{pole})$, where $\mu_\Lambda^\chi = \sum_{p,m} \chi(p^m)(\log p)p^{-m/2} \delta_{m \log p}$ is the **character-decorated prime-orbit measure**. This identifies the singular part of the screw kernel with an orbit measure and thereby states, exactly, what a dynamical realisation would have to produce: orbits carrying a Galois holonomy (POME-K, §7.3, OPEN).

(iii) A certified Weil audit. The arithmetic side of the Weil functional is enclosed in rigorous intervals by ball arithmetic, with the prime tail bounded in closed form (Abel summation against $\psi(x) < 1.03883x$) and the gamma-factor tail by a two-term Watson expansion. On the canonical 12-point grid inherited from earlier work, **0 of 48 channel entries is certifiably negative and the V_4 sum is CERTIFIED POSITIVE at all twelve grid points, without assuming GRH.** Historical negative readings on that grid are refuted. Positivity is further certified, by an interval Cholesky factorisation, on an **11-dimensional space** of test functions.

(iv) Character-channel preconditioning — a certified case study. Under GRH the Weil functional is a **sum of squares** over the zeros, $W_\chi(v * \tilde{v}) = \sum_\gamma |\hat{v}(\gamma)|^2$. Retaining a **single** term therefore gives a rigorous lower bound, and for a Gaussian test packet of width a centred at t the retained term is governed by $\exp(-(\gamma_{\text{cert}} - t)^2/2a)$, where γ is any certified zero of the channel. For ζ , $\gamma_{\text{cert}} = 14.1347$ and the certified lower bound at $(a,t) = (0.2, 0)$ is 1.28×10^{-216} ; the closed-form prime-tail bound of §4 then guarantees a certified **sign** only after a cutoff of about 10^{32} . But K carries a channel, χ_{-11} , whose lowest zero is **2.477**, and the same one-term bound gives 5.83×10^{-6} — certified after a cutoff of **1.5×10^6** .

****Certified prime budget at $(a, t) = (0.2, 0)$: ζ alone $\approx 10^{32}$; the decorated audit $\approx 10^6$. A collapse of 25.4 orders of magnitude.****

The sharpest form of the statement uses different currencies on the two sides. At the cutoff actually used, $P = 4 \times 10^6$, the ζ enclosure at $(0.2, 0)$ is **UNDETERMINED** — and even **granting GRH** this method would need $\approx 10^{32}$ primes to decide its sign. At the same cutoff the V_4 -decorated functional is **CERTIFIED POSITIVE with no arithmetic hypothesis at all. The decorated audit decides a sign the undecorated one cannot.**

And the effect obeys a law. Equating the closed-form tail with the one-term target gives **** $\log N^*(a,t,\gamma) = |\gamma - t|/a + 1/(2a) + \dots$ **** (Theorem T9; accurate to under one nat while $\log N^*$ ranges over 2.8–97.5). **The cost of a certified sign is linear in the distance from the test frequency to the nearest usable zero, divided by the Gaussian width.** The 25.4 orders are then not an accident of this field — they are $(14.1347 - 2.4772)/0.2$ nats, the law evaluated at K 's lowest certified zero.

The effect is **not** a property of the conductor — the lowest zero of $L(\cdot, \chi)$ is not monotone in q (6.02 at $q = 4$, 6.65 at $q = 5$; 2.48 at $q = 11$, 3.12 at $q = 13$). It is a property of **whether the character group contains a low-zero channel**. That is the correct and provable statement, and it is what makes the audit of §4–§5 computable at all. Whether enlarging a character family reliably supplies such a channel is left as a heuristic and an open problem (§6.7).

Nothing here proves or advances RH or GRH. **Global** positivity of the Weil functional on the full admissible test space is equivalent to GRH(ζ_K); the finite-grid and finite-dimensional certificates established here are unconditional but do **not** imply it. §7 records the open operator frontier honestly, including a conditional lower

bound on the assembly error of Suzuki’s finite-interval operator and the precise statement — the Weyl–truncation bridge — that would close the Herglotz route.

Keywords: screw function, Krein–Langer, extended Selberg class, Dedekind zeta, genus theory, Frobenius detector, Weil quadratic form, interval arithmetic, ball arithmetic, certification budget, preconditioning, prime-orbit measure.

§0.1 Epistemic status legend

STATUS	MEANING
PROVEN	Complete proof here; only classical theory as input.
IMPORTED	Proved externally; used without re-proof; cited.
DERIVED	Consequence of PROVEN/IMPORTED items; zero free parameters.
DERIVED-CONDITIONAL	Derived under a stated hypothesis (usually RH/GRH).
CERTIFIED	Numerical statement decided inside ball arithmetic with all truncations bounded in closed form.
VERIFIED	Numerical statement to a declared precision, with the error budget stated but not enclosed.
OPEN	Recognised gap with an explicit closure condition.
NON-CLAIM	Explicit statement of what is not established.

§1. Introduction, contributions, and scope

Imported machinery. (I1) [2, Thm 1.1]: for F in the extended Selberg class with no real zeros except possibly at $\frac{1}{2}$, $\text{GRH}(F) \Leftrightarrow \Re(-g_F(t)) \geq 0$ for $t \geq t_0$. (I2) [2, Thm 4.1]: the zero-free closed form of g_F . (I3) Krein–Langer [4]: g is a screw function iff the kernel $K_g(t,u) = g(t-u) - g(t) - g(-u) + g(0)$ is positive semidefinite. (I4) Schoenberg [5]: ψ with $\psi(0) = 0$ is conditionally negative definite iff $\exp(-\tau\psi)$ is positive definite for all $\tau > 0$. (I5) [3]: the finite-interval operator $A_a = \text{Friedrichs}(D^*G_aD)$ is self-adjoint, lower semibounded, with discrete spectrum; its $a \rightarrow \infty$ limit is left conjectural.

****Sign conventions.** $K_g \ni 0 \Leftrightarrow$ **** g is conditionally positive definite**** $\Leftrightarrow -g$ is conditionally negative definite $\Leftrightarrow \exp(\tau g)$ is positive definite for every $\tau > 0$. Under GRH, $-g(t) = \sum_\gamma (1 - \cos \gamma t)/\gamma^2$.

§1.1 Why this field

$K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$ is a small imaginary biquadratic field whose four genus characters have **pairwise distinct and small** conductors 1, 3, 11, 33 (disc $K = 1089 = 33^2$, and $1 \cdot 3 \cdot 11 \cdot 33 = 1089$). It is therefore a compact four-channel laboratory in which the abelian decomposition of §2 is non-trivial in every channel, and — as §6 shows — the conductor 11 supplies a channel which carries a certified low-lying zero at 2.477, which is what makes the whole audit computable.

§1.2 Contributions and status

#	Result	Status
T1	Abelian screw decomposition $g_K = \sum_{\chi \in \hat{G}} g_\chi$ and the Frobenius coefficient layer $\sum_\chi \chi(\text{Frob}) = G \cdot 1[\text{Frob} = e]$	PROVEN
T2	Channel kernels with $\beta_\chi = (L'/L)^{(\frac{1}{2}, \chi)}$ forced by the functional equation	DERIVED (specialisation of I2)
T3	No real zeros of $L(\sigma, \chi)$ on $(0,1)$ for the three non-principal channels — the standing hypothesis of I1	CERTIFIED
T4	Kink-measure form: $g_\chi'' = \mu_\chi \wedge \chi + 2\beta_\chi \delta_0 + T_\chi\{c_\chi\} - (\text{pole})$	PROVEN + VERIFIED
T5	Rigorous enclosures of the Weil functional; 0/48 negative; V_4 sum 12/12 positive **without GRH**	CERTIFIED
T6	Weil form positive definite on an 11-dimensional test-function space (interval Cholesky)	CERTIFIED
T7	One-term bound: $W_\chi(F_{\{a,t\}}) \geq 2 \hat{v}(\gamma_{\text{cert}}(\chi)) ^2$, hence $W_K \geq 2 \hat{v}(\gamma_{\{\text{cert}, \min\}}(K)) ^2$ — the budget is set by the smallest certified $ \gamma - t $ across channels	DERIVED-CONDIT IONAL on GRH

#	Result	Status
T9	**Certification threshold law** : $\log N^* = d/a + 1/(2a) + (1/d)\log(A/B) + 1/d^2 + O(1/d^3)$, $d = \gamma - t $ — the cost of a certified sign is linear in the distance to the nearest usable zero over the Gaussian width; accurate to 2×10^{-4} nats where $\log N^* \approx 73$	DERIVED-ASYMP TOTIC ($d^2/a \geq 30$) + VERIFIED
T10	**Frobenius-resolved kernel** : $g_h := (1/G)\sum \chi \bar{\chi}(h)g_\chi$ has prime coefficients $\Lambda(n)n^{-\{ -1/2 \}} \cdot 1[\text{Frob} = h]$ at every **unramified** p^m — a Chebotarev resolution of the kernel	PROVEN
T11	**Channel Perturbation Theorem** : h_χ is a finite signed measure on $[-2a, 2a]$ (the archimedean singularity cancels; the δ_0 mass is $-\log q_\chi$), so $q_\chi = q_\zeta + (L^2\text{-bounded form})$. Suzuki's ζ operator theorem therefore transfers to every **real primitive** Dirichlet channel (and hence to multiquadratic abelian Dedekind zetas; complex conjugate-pair channels need a separate realified block), with $ \lambda_n(A_a^\chi) - \lambda_n(A_a^\zeta) \leq \ v_{\chi,a}\ _{TV}$	**PROVEN**
T12	**The Galois-graded operator** $\mathfrak{A}_a$ on $L^2(-a,a) \otimes \mathbb{C}[G]$: self-adjoint, lower semibounded, discrete, G-equivariant, $\hat{G}$ -graded — unconditional, given T11	DERIVED
O6	POME-K and its pre-gate (§8.4): the unpruned map has infinitely many periodic points (DERIVED); the entropy and orbit-count conclusions are **OPEN**	OPEN
T8	Certified case study: at $(0.2, 0)$ the ζ enclosure is undetermined and needs $\approx 10^{32}$ primes even granting GRH, while the decorated enclosure is certified positive **unconditionally** at an explicit cutoff $P = 4 \times 10^6$ plus analytic tail — a 25.4-order collapse of the budget	CERTIFIED
O1	POME-K: a dynamical realisation of g_K must carry a Galois holonomy on its orbits (§7.3)	OPEN (pre-registered)
O4	The family statement: does enlarging a character group reliably supply a low-zero channel? (§6.4)	OPEN / HEURISTIC
O2	The Weyl-truncation bridge for the Herglotz route to RH	OPEN (load-bearing)
O3	A rigorous ball-arithmetic assembly of Suzuki's A_a (the operator-norm error of a truncated assembly)	OPEN

§1.3 Scope

This paper does not prove or advance RH or GRH (NC-1). It consumes $\Lambda(n)\chi(n)$ as external arithmetic input and derives it from nothing (NC-2). The field K and its V_4 data are **given arithmetic input**; no derivation of K from any external structure is claimed (NC-3). The certified positivity results concern a finite grid and a finite-dimensional subspace; Weil positivity is the infinite-dimensional limit and is exactly $\text{GRH}(\zeta_K)$ (NC-4). §7 lists what remains open, with closure conditions.

§2. The abelian screw decomposition

§2.1 Channel kernels

Theorem T2 [DERIVED]. For χ primitive of conductor q_χ and parity a_χ , with $c_\chi = 1/4 + a_\chi/2$,

$$g_\chi(t) = \sum_{\{n \leq e^{\{t\}}\}} \chi(n) \Lambda(n) n^{\{ -1/2 \}} (|t| - \log n) + \beta_\chi |t| - 1/4 [\Phi(1, 2, c_\chi) - e^{\{ -2c_\chi |t| \}} \Phi(e^{\{ -2|t| \}}, 2, c_\chi)],$$

$$\beta_\chi = -1/2 (\psi(c_\chi) - \log(\pi/q_\chi)),$$

**** β_χ is defined by the gamma-factor expression above****, which makes sense for every primitive χ . ****When $L(1/2, \chi) \neq 0$ **** — which §4.3 certifies for the three non-principal channels of K — the functional equation identifies it with $(L'/L)(1/2, \chi)$. Theorem T11 uses only the gamma-factor definition and therefore does not require a non-vanishing central value; the identification with the logarithmic derivative is used only where it is certified.

with the extra term $-4(e^{\{t/2\}} + e^{\{-t/2\}} - 2)$ when $\chi = 1$. The coefficient β_χ is **not a convention**: differentiating $\Lambda(s, \chi) = \varepsilon(\chi) \Lambda(1-s, \chi)$ at $s = 1/2$ with $\varepsilon = +1$ gives $\Lambda'(1/2, \chi) = 0$, hence $(L'/L)(1/2, \chi) = -1/2 (\psi(c_\chi) - \log(\pi/q_\chi)) = \beta_\chi$. [Certified against numerical logarithmic differentiation to 5×10^{-17} .]

Table 2.1. The four channels of K . All $\varepsilon(\chi) = +1$, so $\Lambda(1/2 + it, \chi)$ is real (verified to 1.5×10^{-15}).

χ	q_χ	a_χ	c_χ	$\beta_\chi = (L'/L)(1/2, \chi)$	certified low-lying zero γ_{cert}
1 (ζ)	1	0	1/4	2.6860917096	14.134725
χ_{-3}	3	1	3/4	0.5659892385	8.039737
χ_{-11}	11	1	3/4	-0.0836522536	**2.477244**
χ_{33}	33	0	1/4	0.9378379289	2.996951

The bold entry is the pivot of the whole paper: see §6.

§2.2 The decomposition and the Frobenius detector

Theorem T1 [PROVEN]. Let $K/\mathbb{Q}$ be finite abelian with group G and character group $\hat{G}$ (viewed as primitive Dirichlet characters). Then $\xi_K \in S\# \mathfrak{b}$ and

$$g_K(t) = \sum_{\chi \in \hat{G}} g_{\chi}(t); \text{ and for } p \text{ unramified, } \sum_{\chi \in \hat{G}} \chi(p^m) = |G| \cdot 1[\text{Frob}_p(p^m) = e].$$

Proof. $\zeta_K = \prod_{\chi} L(\cdot, \chi)$; $S\#$ is a multiplicative monoid, so ξ'/ξ is additive over the factorisation, and by [2, (3.1)] the transform $\int_0^\infty g_F(t) e^{\{izt\}} dt = z^{-2} (\xi'_F/\xi_F)(\frac{1}{2} - iz)$ is additive; uniqueness of the inverse transform gives the first claim. The second is orthogonality of $\hat{G}$.

Table 2.2. Verified coefficient layer for K ($|G| = 4$). Unramified values lie in $\{0, 4\}$ for all $n < 2000$ coprime to 33; $\sum \chi(3^k) = 2$ for all k ; $\sum \chi(11^k) = 2 \cdot 1[k \text{ even}]$ — the fingerprint of $e = f = 2$ at 11.

p	2	3	5	7	11	13	31	37
$\sum \chi(p)$	0	2	0	0	0	0	4	4
$\sum \chi(p^2)$	4	2	4	4	2	4	4	4

§3. The kink measure and the decorated prime-orbit target

Theorem T4 [PROVEN + VERIFIED]. On $\mathbb{R}$, as distributions,

$$g_{\chi}'' = \mu_{\Lambda}^{\chi} + 2\beta_{\chi} \delta_0 + T_{\{c_{\chi}\}} - m_{\chi}(e^{\{|x|/2\}} + e^{\{-|x|/2\}}), \quad \mu_{\Lambda}^{\chi} := \sum_{p, m \geq 1} \chi(p^m) (\log p)^{-m/2} (\delta_{\{m \log p\}} + \delta_{\{-m \log p\}}),$$

where T_c is **defined as the distribution** $D^2[-\frac{1}{4}\sum_{k \geq 0} (1 - e^{\{-2(k+c)|x|\}})/(k+c)^2]$, with $T_c|_{\{\mathbb{R} \setminus \{0\}\}} = e^{\{-2c|x|\}}/(1 - e^{\{-2|x|\}}) \sim 1/(2|x|)$ — which is **not locally integrable at 0**, so T_c and not ω_c is the correct object. Proof: $(|x| - L)_+'' = \delta_{-L} + \delta_{\{-L\}}$; $|x|'' = 2\delta_0$; and the Lerch series differentiates term by term. [VERIFIED: the jump of g_{χ}' at $\log n$ equals $\chi(n)\Lambda(n)n^{-\frac{1}{2}}$ to 8×10^{-6} at ten kinks per channel.]

This is the Weil explicit formula written **on the kernel** rather than on test functions, and it is the form in which the kernel meets a dynamical system: the singular part of g_{χ}'' is a weighted orbit measure.

The kink-measure identity also supplies a precise ****decorated orbit-measure target****: any dynamical system whose Ruelle zeta reproduces g_K must have primitive orbits carrying a Galois holonomy $h_{\chi} \in \text{Gal}(K/\mathbb{Q})$ with $\sum_{\gamma, m} \chi(h_{\chi}^m) \ell_{\gamma} e^{\{-m \ell_{\gamma}/2\}} \delta_{\{m \ell_{\gamma}\}} = \mu_{\Lambda}^{\chi}$ for every χ . Its dynamical realisation is left open and is discussed in the Outlook (§7.3).

§4. The certified Weil audit

§4.1 The two sides, and the two tails

For $v(x) = e^{\{-ax^2\}} \cos(tx)$ and $F = v * \tilde{v}$, integration by parts against the screw kernel gives exactly

$$W_{\chi}(F) = -\int_{\mathbb{R}} g_{\chi}(x) F''(x) dx = \sum_{\gamma} |\hat{v}(\gamma)|^2 \quad (\text{the zero side, a sum of squares under GRH}).$$

The arithmetic side is enclosed in ball arithmetic; the only two truncations are bounded in closed form.

(R1) Prime tail. With $G(u) = C(1+\kappa)e^{\{-au^2/2\}} \geq |F(u)|$, Abel summation against $\psi(x) < 1.03883x$ (Rosser–Schoenfeld) and completion of the square give

$$|E_P| \leq 2 \cdot 1.03883 \cdot C(1+\kappa) e^{\{1/(8a)\}} [e^{\{-as_0^2/2\}} + \sqrt{(\pi/2a)} \cdot \text{erfc}(s_0 \sqrt{(a/2)})], \quad s_0 = \log N - 1/(2a).$$

(R2) Gamma-factor tail. F is entire and even, so $F'(0) = F'''(0) = F^{(5)}(0) = 0$ and Watson's lemma with explicit remainder gives

$$|\frac{1}{2} \sum_{k \geq K} J(2\mu_k)/\mu_k^2 - \frac{1}{2} [F''(0)\zeta(3, K+c)/2 + F^{(4)}(0)\zeta(5, K+c)/8]| \leq \|F^{(6)}\|_{\infty} \cdot \zeta(7, K+c)/64,$$

with $\|F^{(6)}\|_{\infty} \leq (1/2\pi) \int_{\mathbb{R}} \hat{F}(r) dr$ — a Gaussian sixth moment, legitimate because $\hat{F} = |\hat{v}|^2 \geq 0$. Both remainders are controlled by explicit analytic bounds; **no numerical quadrature is used for either tail**.

§4.2 Theorem T5 [CERTIFIED]

On the canonical grid $(a, t) \in \{0.2, 0.5, 1.0\} \times \{0, 1, 5, 14.13\}$, the interval $[L, U]$ provably contains

$W_{\chi}(F_{\{a, t\}})$. **The certificate sums prime powers explicitly through $P = 4 \times 10^6$ and encloses the remainder**

analytically by the closed-form tail (R1), whose value at $a = 0.2$ is then $\approx 1.1 \times 10^{-6}$, comfortably below the V_4 target of 6.8×10^{-6} . All statements below refer to that protocol: explicit summation to 4×10^6 plus a rigorous analytic tail.) for each of the 48 entries. **No entry is certifiably negative; 33 are certifiably positive; and the V_4 sum $\Sigma_{\chi} W_{\chi}$ is certifiably positive at all twelve grid points.** No arithmetic hypothesis is used — in particular GRH is not assumed.

Table 4.1. The V_4 -summed Weil functional on the canonical grid (arb, 320 bits). All twelve certified positive.

(a, t)	L	U	verdict
(0.2, 0)	6.182e-6	7.573e-6	POSITIVE
(0.2, 1)	3.392e-2	3.392e-2	POSITIVE
(0.2, 5)	3.859e+0	3.859e+0	POSITIVE
(0.2, 14.13)	1.520e+1	1.520e+1	POSITIVE
(0.5, 0)	2.875e-2	2.875e-2	POSITIVE
(0.5, 1)	4.179e-1	4.179e-1	POSITIVE
(0.5, 5)	3.023e+0	3.023e+0	POSITIVE
(0.5, 14.13)	9.217e+0	9.217e+0	POSITIVE
(1.0, 0)	3.629e-1	3.629e-1	POSITIVE
(1.0, 1)	8.597e-1	8.597e-1	POSITIVE
(1.0, 5)	2.613e+0	2.613e+0	POSITIVE
(1.0, 14.13)	6.692e+0	6.692e+0	POSITIVE

Historical readings of ‘negative’ entries on this grid (5/12, 4/12, 1/12 in earlier internal work) are ****refuted****: they were the sign of a truncation error. The complete 48-entry table and the per-channel verdicts are in the Supplement.

§4.3 Theorem T3 [CERTIFIED] — the standing hypothesis of (I1)

(I1) requires that $L(\cdot, \chi)$ have no real zeros in $(0, 1)$. The functional equation with a positive gamma factor reduces this to $L > 0$ on $[\frac{1}{2}, 1]$. On $[\frac{1}{2}, 0.9]$ we certify by adaptively subdivided Hurwitz-zeta balls; near $s = 1$ that representation is useless to interval arithmetic (the poles cancel only because $\Sigma \chi(r) = 0$, a cancellation an interval evaluation cannot see), so on $[0.9, 1]$ we use a truncated Dirichlet series with the rigorous Abel remainder $|\Sigma_{n > M} \chi(n) n^{-\sigma}| \leq 2B_{\chi} M^{-\sigma}$, $B_{\chi} = \max_x |\Sigma_{n \leq x} \chi(n)| = 1, 3, 3$. Every enclosure is strictly positive.

§5. Positivity on a finite-dimensional space

A grid of twelve test functions is twelve points. The natural strengthening is a **space**. With $v_j(x) = e^{-ax^2} \cos(jx)$, $a = \frac{1}{2}$, $V_N = \text{span}\{v_0, \dots, v_{N-1}\}$, the Gram entries have the closed form

$$v_j * \tilde{v}_k = A_1 e^{-ax^2/2} \cos(\omega_1 x) + A_2 e^{-ax^2/2} \cos(\omega_2 x), \quad \omega_{\{1,2\}} = (t_j \pm t_k)/2, \quad A_{\{1,2\}} = \frac{1}{2} \sqrt{(\pi/2a)} e^{-\{(t_j \mp t_k)^2/(8a)\}},$$

so every entry of M_N is two evaluations of the same shape functional that §4 already encloses.

Theorem T6 [CERTIFIED]. With $P_{\max} = 2 \times 10^6$ and $K = 6000$, the V_4 Weil form is positive definite on V_N for every $N \leq 11$. The certificate is an **interval Cholesky**: with L the Cholesky factor of $\text{mid}(M_N)$ (80 digits, then treated as exact), $E := L^{-1} M_N L^{-T} - I$ is computed in ball arithmetic and $\|E\|_F < 1$ is certified; then $M_N = L(I+E)L^T > 0$.

Table 5.1. Interval Cholesky certificate. The $\lambda_{\min}$ column is descriptive (a high-precision midpoint eigenvalue); the certificate is $\|E\|_F$.

N	$\lambda_{\min}(\text{mid } M_N)$	certified $\ E\ _F$	verdict
4	1.572e-09	6.80e-19	CERTIFIED $M_N > 0$
6	6.322e-13	3.11e-14	CERTIFIED $M_N > 0$
8	2.220e-15	2.45e-04	CERTIFIED $M_N > 0$
10	1.166e-17	4.68e-02	CERTIFIED $M_N > 0$
11	8.674e-19	6.28e-01	CERTIFIED $M_N > 0$
12	—	1.67e+00	not certified

****Scope.**** V_N is the fixed-width family $a = 1/2$ with integer frequencies; it contains three of the twelve canonical test functions, namely $(a,t) = (0.5, 0), (0.5, 1), (0.5, 5)$. What is certified is positivity on an entire 11-dimensional space, i.e. on an uncountable family, not at isolated points.

§6. Character-channel preconditioning

This section is the new contribution. It is independent of the operator-theoretic programme; it is a statement about **how to compute**; and it is what makes §4 and §5 possible.

§6.1 One term of a sum of squares

Under GRH the Weil functional is a Gram evaluation against the zeros, $W_\chi(v * \hat{v}) = \sum_\gamma |\hat{v}(\gamma)|^2$. Every summand is a **square**, so retaining any single zero gives a rigorous lower bound. Retaining the lowest:

****Theorem T7 (one-term bound) [DERIVED-CONDITIONAL on GRH].**** $W_\chi(F_{\{a,t\}}) \geq 2 |\hat{v}_{\{a,t\}}(\gamma)|^2$ for ****any**** zero γ of the channel $= (\pi/2a) \cdot [e^{\{-(\gamma-t)^2/4a\}} + e^{\{-(\gamma+t)^2/4a\}}]^2$,

and for a finite abelian K with character group $\hat{G}$, since $W_K = \sum_\chi W_\chi$ with every term non-negative,

$$W_K(F_{\{a,t\}}) \geq 2 |\hat{v}_{\{a,t\}}(\gamma_{\text{low}})|^2 \text{ for **any** certified zero } \gamma_{\text{low}} \text{ of **any** channel of } K.$$

The budget is therefore set by one number: the smallest $|\gamma - t|$ among the certified zeros of all channels. The bound is computed inside ball arithmetic: γ is enclosed by a strict sign change of $\Lambda(1/2 + i\gamma, \chi)$ in arb, and $|\hat{v}|^2$ is bounded below on the bracket by monotonicity of each Gaussian in the distance from its centre.

****Firstness is not claimed.**** A sign change certifies a zero of ****odd multiplicity**** in the bracket; it does not exclude a lower zero of even multiplicity, nor an undetected pair. ****Nothing in T7–T9 needs firstness****: the zero side is a sum of non-negative terms, so ****one certified low-lying zero suffices****. A genuine first-zero claim would need a Turing-type zero count, which is not run here; the OPEN registry says so.

§6.2 Theorem T9 — the certification threshold law

T7 says the budget is set by one number. **T9 says exactly which number, and how.** Equate the closed-form tail (R1) with the one-term target of T7. Writing $s_0 = \log N - 1/(2a)$, the tail is $A(a,t) \cdot e^{\{-a s_0^2/2\}} (1 + O(1/(a s_0)))$ with $A(a,t) = 2 \cdot 1.03883 \cdot C(1+\kappa) e^{\{1/(8a)\}}$, and the target is $(\pi/2a) \cdot e^{\{-(\gamma-t)^2/(2a)\}} (1 + o(1))$. Solving for s_0 :

****Theorem T9 (Certification Threshold Law) [DERIVED-ASYMPTOTIC in $d^2/a \rightarrow \infty$; VERIFIED].**** $\log N^*(a, t, \gamma) = **d/a** + 1/(2a) + (1/d) \cdot \log(A(a,t)/B(a,t,\gamma)) + **1/d^2** - (1 + aL^2)/(2d^3) + O_{\{a,t\}}(d^{-4})$, $d := |\gamma - t|$, $L := \log(A/B)$,

with $A(a,t) = 2 \cdot 1.03883 \cdot C(1+\kappa) e^{\{1/(8a)\}}$ the tail prefactor and $B(a,t,\gamma) = (\pi/2a) \cdot (1 + e^{\{-\gamma t/a\}})^2$ the target prefactor — note the factor $(1 + e^{\{-\gamma t/a\}})^2$, which equals **4** at $t = 0$ because the two Gaussian lobes of $\hat{v}$ coincide there.

The cost of a certified sign is linear in the distance from the test frequency to the nearest usable zero, divided by the Gaussian width. The subleading term is a single logarithm over d . The expansion is asymptotic in d^2/a ; near resonance ($d^2/a \lesssim 1$) it breaks down and the tail bound must be inverted directly, which is what the code does.

The $1/d^2$ term is derived, not fitted. The erfc part of the tail contributes a factor $[1 + 1/(a s_0) - 1/(a^2 s_0^3) + \dots]$; substituting $s_0 = d/a + \delta$ and expanding gives $d\delta + (a/2)\delta^2 = L + 1/d - 1/(2d^2) + O(d^{-3})$ with $L := \log(A/B)$, hence

$$\delta = L/d + 1/d^2 - (1 + aL^2)/(2d^3) + O(d^{-4}).$$

****Correction to the previous version.**** It claimed ‘the remainder is independent of a ’. That is false, and is ****retracted****: only the ****universal $1/d^2$ coefficient**** is independent of a . The next coefficient, $-(1 + aL^2)/2$, depends on both a and L . What the a -independence of the $1/d^2$ term does explain is the striking constancy of the residual across $a = 0.15 \dots 1.0$ in Table 6.1.

****Accuracy (Table 6.1). **** With the $1/d^2$ term the residual is $O(1/d^3)$. At $\gamma = 14.13$ the law is accurate to $**2 \times 10^{-4}$ nats** while $\log N^*$ runs from 14.6 to 97.5 — a relative accuracy of 2×10^{-6} — and this holds unchanged for $a = 0.15, 0.2, 0.3, 0.5, 0.8, 1.0$. At $\gamma = 8.04$ it is 1×10^{-3} ; at $\gamma = 2.48$ ($d^2/a \approx 31$, the edge of the regime) it is 0.03. Below $d^2/a \approx 30$ the expansion is not used: the tail bound is inverted directly in arb.

Table 6.1. Theorem T9 (closed form, with the derived $1/d^2$ term) against the exact cutoff obtained by inverting the tail bound in arb. ‘v2.3’ is the law without the $1/d^2$ term, shown for contrast.

(a, t)	γ	exact log N*	d^2/a	law error (no $1/d^2$)	**law error (T9)**
(0.15, 0)	14.135	97.499	1332	-0.0048	**+0.00019**
(0.20, 0)	14.135	73.103	999	-0.0048	**+0.00020**
(0.30, 0)	14.135	48.712	666	-0.0048	**+0.00022**
(0.50, 0)	14.135	29.205	400	-0.0048	**+0.00024**
(1.00, 0)	14.135	14.586	200	-0.0048	**+0.00026**
(0.20, 0)	8.040	42.581	323	-0.0144	+0.0011
(0.50, 1)	8.040	15.102	99	-0.0187	+0.0015
(0.20, 0)	2.477	14.591	31	-0.1325	+0.0305

Corollary T9a (the preconditioning gain, in closed form) [DERIVED]. Replacing the ζ channel’s usable zero by a low-lying zero of a decorated channel changes the certified budget by

$$\Delta \log N^* = (|\gamma_{\zeta} - t| - |\gamma_{\text{low}} - t|) / a + O(a \cdot \log(\cdot) / |\gamma|),$$

which at $(a, t) = (0.2, 0)$ with $\gamma_{\zeta} = 14.1347$ and $\gamma_{\text{low}} = 2.4772$ gives $(14.1347 - 2.4772)/0.2 = \mathbf{58.29 \text{ nats} = 25.31 \text{ decimal orders}}$ — matching the certified value of 25.4 in Table 6.2. **The 25 orders are not a coincidence of this field: they are the law evaluated at this field’s lowest certified zero.**

****Why this is the theoretical content, and not merely a computation.**** T9 turns the question ‘can this Weil functional be audited?’ into an arithmetic one: *how far is the nearest zero, in units of the test width?* It predicts the cost before any computation is run; it explains why the ζ audit at low frequency is hopeless ($\gamma_{\text{cert}}/a = 70$ nats) and why the decorated audit is routine ($\gamma_{\text{low}}/a = 12$ nats); and it converts the choice of field into an optimisation — minimise $|\gamma - t|$ over the channels. To our knowledge no such law has been stated.

§6.3 Theorem T8 [CERTIFIED case study] — the budget collapse for K

Inverting the closed-form prime tail (R1) of §4.1 gives, for a target τ , the cutoff $N^*(a, t, \tau)$ at which $|E_P| < \tau$ — the **certification budget**: the price, in primes, that *this method* must pay for a legible sign. Applying it to the one-term bounds of §6.1, entirely in ball arithmetic:

Table 6.2. Certified zero brackets (strict ball sign changes of $\Lambda(\frac{1}{2} + i\gamma, \chi)$) and the resulting certified budgets. the smallest certified sign-change zero found is 2.477, supplied by χ_{-11} . The last column uses the ****unconditional**** arithmetic enclosure of §4.2 as the decorated target — no GRH — against the GRH-conditional one-term bound of the ζ channel. Every entry is decided in arb (blocks D0–D6, 16/16 PASS).

χ	certified γ bracket	(a,t)	one-term LB (GRH)	log N*	uncond. V_4 target $\rightarrow$ log N*	gain over ζ
ζ	[14.130, 14.150]	(0.2, 0)	1.283e-216	73.2	—	—
χ_{-11} / V_4	[2.470, 2.490]	(0.2, 0)	5.827e-06	14.7	6.182e-06 $\rightarrow$ 14.6	**25.4 orders**
ζ	[14.130, 14.150]	(0.2, 1)	1.402e-187	68.2	—	—
χ_{-11} / V_4	[2.470, 2.490]	(0.2, 1)	3.052e-02	10.1	3.392e-02 $\rightarrow$ 10.0	**25.3 orders**
ζ	[14.130, 14.150]	(0.5, 0)	1.392e-86	29.2	—	—
χ_{-11} / V_4	[2.470, 2.490]	(0.5, 0)	2.550e-02	5.7	2.875e-02 $\rightarrow$ 5.7	**10.2 orders**
ζ	[14.130, 14.150]	(1.0, 0)	2.091e-43	14.6	—	—
χ_{-11} / V_4	[2.470, 2.490]	(1.0, 0)	2.830e-01	5.0	3.629e-01 $\rightarrow$ 5.0	4.2 orders

§6.4 The unconditional form of the comparison — and why it is the strongest one

Theorem T7 is GRH-conditional on both sides, which makes the comparison fair but not maximally sharp. The sharp form uses **different currencies on the two sides**, and it is the honest one:

- On the ζ side we **grant GRH**. Even then, the one-term bound is 1.283×10^{-216} and the present tail bound certifies the sign only after a cutoff of $\approx 10^{32}$. At the cutoff actually used, $P = 4 \times 10^6$, the ζ enclosure at $(0.2, 0)$ is **UNDETERMINED**: the interval contains zero.

• On the decorated side we **grant nothing**. The arithmetic enclosure of §4.2 is unconditional — it uses only the explicit formula, the Chebyshev bound $\psi(x) < 1.03883x$, and ball arithmetic — and at the same cutoff it gives $[6.182 \times 10^{-6}, 7.573 \times 10^{-6}]$: **CERTIFIED POSITIVE, without GRH**.

Theorem T8 (unconditional form) [CERTIFIED]. At $(a, t) = (0.2, 0)$ and $P = 4 \times 10^6$: the ζ channel's sign is not decided, and — granting GRH — this method would need about 10^{32} primes to decide it; the V_4 -decorated functional's sign is decided, positively, with no arithmetic hypothesis whatever. **The decorated audit sees a sign that the undecorated one cannot.**

This is the form to quote. It does not compare two conditional bounds; it compares *what can be certified* with *what cannot*, at the same computational cost, on the same grid point.

§6.5 What the table does NOT say

****Two statements this table does NOT make.**** (a) 1.283×10^{-216} is ****not**** the true value of W_ζ at $(0.2, 0)$; it is a certified GRH-conditional ****lower bound**** — the omitted zeros contribute positively. (b) 10^{32} is ****not**** an information-theoretic minimum; it is the cutoff at which the ****present closed-form tail bound**** guarantees a certified sign. A sharper tail bound, or a different method, could do better. What the table does say is that with ***this*** protocol, and on this grid, the decorated audit is certified 25 orders of magnitude more cheaply.

§6.6 Why the historical negatives happened

Corollary [DERIVED]. Earlier computations on this grid used $P_{\max} = 500$. At $a = 0.2$ the closed-form prime-tail bound at that cutoff is of order 10^{-1} , while the signal at the worst grid point is below 10^{-216} . **The sign that was read was the sign of the truncation error.** The remedy is not more primes — no attainable cutoff repairs the ζ channel at low frequency — it is the low-zero channel.

****The one-sentence export.**** *A negative truncated Weil or screw computation is scientifically meaningless unless it comes with an error enclosure smaller than the target margin; at low frequency on ζ no attainable enclosure is; and a character decomposition containing a low-zero channel supplies one.*

§6.7 The generalisation is open, and the obvious guess is wrong

It is tempting to say that a larger conductor gives a lower first zero (i.e. a lower γ_{cert}), since the zero density of $L(\cdot, \chi)$ is $(1/2\pi)\log(qT/2\pi e)$. **That is false as stated.** The smallest certified sign-change zeros found for the real primitive characters of small conductor are

q	1 (ζ)	3	4	5	7	11	13	33
γ_{cert}	14.1347	8.0397	6.0210	6.6485	4.4757	**2.4772**	3.1193	2.9970

— not monotone (6.02 at $q = 4$ but 6.65 at $q = 5$; 2.48 at $q = 11$ but 3.12 at $q = 13$). The density heuristic controls an **average**, not the first zero of an individual L-function.

Conjectural design principle [HEURISTIC / OPEN]. A character decomposition containing a low-zero channel acts as a numerical preconditioner for a finite Weil audit; and since γ_{cert} is, empirically, a roughly independent draw per channel, enlarging the character group raises the chance that $\gamma_{\text{cert}}^{\min}$ is small. Quantifying that — e.g. a bound on the expected $\gamma_{\text{cert}}^{\min}$ over the genus characters of K as a function of $|G|$ and disc K — is open. **What is established here is the mechanism (T7) and the case (T8); the family statement is a target, not a theorem.**

§6.8 Two further failure modes of numerical Weil work

(a) The margin collapses with dimension. On the certified range $2 \leq N \leq 10$ of §5, $\log_{10} \lambda_{\min}(M_N) \approx -1.677N - 1.136$ (least squares; descriptive, no asymptotic law claimed). The Weil form is the Gram matrix of the zero-evaluation map, so its **effective numerical rank** is bounded by the number of zeros the subspace resolves in its frequency window. Certification buys dimensions, not the theorem: the tail step is GRH.

(b) A truncated kernel needs all primes to $e^{\{2a\}}$. Any finite-interval construction from g on $[-a, a]$ evaluates the kernel at $|t| \leq 2a$, and the prime part of g at t carries every prime power $n \leq e^{\{t\}}$; so exact assembly needs **all** prime powers to $e^{\{2a\}}$. (The cutoff is already visible in Suzuki's own finite-interval form; the contribution here is to read it as a budget.) A **fixed** shortfall δ in the log-cutoff costs, pointwise at $t = 2a$, $\frac{1}{2} e^{\{a\}} \delta^2 + O(e^{\{a\}} \delta^3)$ — an error that **grows with a** . The corresponding operator-norm bound $\|\Delta A_a(P)\| \leq \int_{-a}^a \log P^{\{2a\}} |\Delta g_P(t)| \cdot \|Q_a(t)\| dt$ has **not** been evaluated (O3). With the prime table cut at e^8 instead of e^{16} , the screw margin of the K-kernel at $a = 8$ returns -29424.7 instead of $+0.204$.

§7. The open operator frontier

The operator programmes in this section remain open; several supporting propositions and no-go results are established. They are recorded because the closure conditions are precise, and because two of them were reached by eliminating wrong answers.

§7.1 Suzuki's finite- a operator: two tracks, not one

[3] proves (I5) that A_a is self-adjoint, lower semibounded, with discrete spectrum. It does **not** prove $A_a \ni 0$ — positivity for all a is RH-equivalent. Two distinct conjectural limits must not be conflated:

Track G (the motivating statement): $c_a \cdot \hat{v}_a(z) \rightarrow \xi(\frac{1}{2} + iz)$ for the ground vector v_a .

Track W (the main conjectural statement of [3]): with boundary data $W(a, \theta; z)$ of the self-adjoint extensions and a normalisation $\phi(a, z)$, $e^{\{a\}} W(a, \theta; z) \rightarrow z^2 \xi'/\xi(\frac{1}{2} - iz)$.

This paper constructs neither. It supplies the budget for constructing them (§6.4(b)) and the following lemma, which is what a pilot computation can legitimately yield.

Lemma 7.1 (conditional assembly-error lower bound) [DERIVED-CONDITIONAL on RH]. Let B^* be the exact Galerkin form matrix of A_a in a basis, B_{num} a computed approximation, and c a normalised coefficient vector. If a zero-side partial sum gives a lower bound $L_T(c) \leq c^* B^* c$ (legitimate under RH, since $c^* B^* c = \sum_{\gamma} |\hat{f}_c(\gamma)|^2 / \gamma^2$ is a sum of squares with $f = Du$), and if $c^* B_{\text{num}} c < 0$, then

$$\|B^* - B_{\text{num}}\|_2 \geq |c^*(B^* - B_{\text{num}})c| \geq L_T(c) - c^* B_{\text{num}} c > 0.$$

This **bounds the assembly error from below**. It does **not** determine $\lambda_{\min}(B^*)$ nor the scale of the true quadratic form: a partial sum of a positive series is a lower bound only, and a Rayleigh quotient at one vector bounds $\lambda_{\min}$ from **above**, not below. [A pilot float64 assembly at $a = 3, 4$ gives $\|\Delta B\|_2 \gtrsim 5.9 \times 10^{-6}$ and $\gtrsim 3.1 \times 10^{-5}$ respectively.

These are diagnostics, not certificates; the numbers and the code are in the Supplement.]

****Two errors are recorded here so they are not repeated.**** (i) A negative computed λ_0 does **not** certify an assembly error, because $A_a \ni 0$ is not an unconditional theorem. (ii) A small zero-side partial sum does **not** show that the true form value is small. Both were asserted in internal drafts and are retracted; see the Supplement.

§7.2 The Herglotz route, and the one statement that would close it

Set $M(z) := i(\xi'/\xi)(\frac{1}{2} - iz)$ on $\mathbb{C}_+$. Four things are proved.

(a) [PROVEN] M is Herglotz on $\mathbb{C}_+$ **iff** RH. ($\Rightarrow$) a Herglotz function is holomorphic, and an off-line zero gives M a pole in $\mathbb{C}_+$. ($\Leftarrow$) under RH the Hadamard product gives $M(z) = \sum_{\gamma > 0} m_{\gamma} [1/(\gamma - z) + 1/(-\gamma - z)]$ (the compensating $-1/\gamma$ terms cancel pairwise, and $\xi'(\frac{1}{2}) = 0$ by evenness), and each summand has positive imaginary part on $\mathbb{C}_+$.

(b) [PROVEN] $|g(t)| \leq 20(1+t)e^{\{t/2\}}$ (Abel + Chebyshev), and for $\Im z > \frac{1}{2}$, $M(z) = i \int_0^{\infty} g(t) e^{\{izt\}} dt$ with $|M(z) - m_a^{\{\text{trunc}\}}(z)| \leq |z|^2 \cdot 20 \cdot e^{\{-(\Im z - \frac{1}{2}) \cdot 2a\}} [(1+2a)/(\Im z - \frac{1}{2}) + 1/(\Im z - \frac{1}{2})^2]$, $m_a^{\{\text{trunc}\}}(z) := iz^2 \int_0^{\{2a\}} g(t) e^{\{izt\}} dt$.

(c) [PROVEN] If (m_a) are Weyl functions of self-adjoint operators (hence Herglotz on all of $\mathbb{C}_+$), $\sup_a |m_a(i)| < \infty$, and $m_a \rightarrow M$ uniformly on **one** compact of $\{\Im z > \frac{1}{2}\}$, then RH. (Herglotz normal families + identity theorem + no poles.)

(d) [PROVEN, negative] m_a^{trunc} is **not** Herglotz — iz^2 times the transform of a compactly supported kernel is entire of exponential type, and a non-affine entire Herglotz function does not exist. So the naive shortcut is closed.

Therefore the whole gate is [OPEN]:

#	Condition	Status
S'1	Fix a canonical Weyl function: A_a alone does not determine m_a — a boundary triple, or a cyclic vector and normalisation, must be chosen.	OPEN
S'2	$\sup_a m_a(i) < \infty$. Since $m_a(i) = \alpha_a + i(\beta_a + \int d\tau_a/(1+t^2))$, this is equivalent to a bound on α_a and on $\beta_a + \int d\tau_a/(1+t^2)$ — not on the measure alone.	OPEN
S'3	The Weyl-truncation bridge: $\sup_{\{z \in K\}} m_a(z) - m_a^{\text{trunc}}(z) \rightarrow 0$ on one compact of $\{z > 1/2\}$. With (b) this is equivalent to $m_a \rightarrow M$.	OPEN — load-bearing

Convergence of Weyl functions is governed by Grommer–Hamburger: $m_a \rightarrow M$ locally uniformly iff $\alpha_a \rightarrow \alpha$, $\tau_a \rightarrow \tau$ vaguely, **and** $\beta_a + \int d\tau_a/(1+t^2) \rightarrow \beta + \int d\tau/(1+t^2)$. Separate convergence of β_a , and weak convergence of the weighted measures, do **not** follow — spectral mass escaping to infinity can be converted into the linear coefficient (take $\tau_n = n^2 \delta_n$, $\alpha_n = \beta_n = 0$: then $m_n \rightarrow z$, whose β is 1).

§7.3 Outlook: the decorated orbit-measure target (POME-K)

Theorem T4 identifies the singular part of g_χ'' with a **character-decorated prime-orbit measure**. That turns a vague programme — ‘realise ζ as a dynamical zeta’ — into an explicit and falsifiable one. Write $\mu_{IT}^\chi = \Sigma_{\{\gamma, m\}} \chi(h_\gamma^m) \ell_\gamma e^{\{-m \ell_\gamma/2\}} \delta_{\{m \ell_\gamma\}}$ for a flow whose primitive orbits carry a holonomy $h_\gamma \in \text{Gal}(K/\mathbb{Q})$. Under local finiteness and absolute convergence of the dynamical Euler product for $\Re s > 1$ with $Z_{IT} \rightarrow 1$ at ∞ ,

POME-K [OPEN, PRE-REGISTERED]: $\mu_{IT}^\chi = \mu_\Lambda^\chi$ for every $\chi \in \hat{G} \Leftrightarrow Z_{IT, \chi}(s) = L(s, \chi)$ for every χ .

(The kernel step: if $D^2 h = 0$ then h is affine; h even kills the linear term; $h(0) = 0$ kills the constant. So equality of the atomic parts on $\mathbb{R} \setminus \{0\}$ forces equality of the kernels, hence of ξ'/ξ .) **The decoration is what gives the target teeth.** A candidate dynamics whose orbit lengths are right but whose holonomy is trivial passes a ζ -only test and fails three of the four decorated ones; a single-channel test cannot see that.

The dynamical realisation of POME-K, and the pre-gate that constrains it, are discussed in §8.4.

§7.4 What is not claimed

NC-1. No progress on RH or GRH. NC-2. $\Lambda(n)\chi(n)$ is external input; nothing here derives the primes. NC-3. K and its V_4 data are given arithmetic input. NC-4. §4 certifies twelve test functions and §5 an 11-dimensional space; Weil positivity is the limit and is exactly GRH(ζ_K). NC-5. Track G and Track W are untested; the pilot assembly of §7.1 has no evidential status. NC-6. POME-K is a target, not a result. NC-7. To our knowledge no certified numerical implementation of Suzuki’s A_a has been published; we claim no more than that.

§8. Frobenius resolution, and a conditional outlook

§8.1 Theorem T10 — the Frobenius-resolved kernel [PROVEN]

The decomposition of §2 splits the kernel by **character**. Inverting the finite Fourier transform on $\hat{G}$ splits it by **Frobenius class**. For $h \in G = \text{Gal}(K/\mathbb{Q})$ define the **Frobenius-resolved kernel component**

$$g_h(t) := (1/|G|) \sum_{\chi \in \hat{G}} \chi(h) \cdot g_\chi(t).$$

Theorem T10 [PROVEN]. For every **unramified** prime power $n = p^m$ ($p \nmid \text{disc } K$), the prime coefficient of g_h at n is

$$\Lambda(n) \cdot n^{-1/2} \cdot \mathbb{I}[\text{Frob}_{\{p^m\}} = h].$$

Proof. The coefficient is $(1/|G|)\Lambda(n)n^{-1/2} \sum \chi \bar{\chi}(h)\chi(\text{Frob}_{\{p^m\}})$, and orthogonality of $\hat{G}$ gives $|G| \cdot 1[\text{Frob} = h]$. [VERIFIED: zero mismatches for every $n < 400$ coprime to 33 and every $h \in V_4$. The ramified primes 3 and 11 are excluded, as they must be: Frobenius is defined only at unramified places.]

****What T10 is, precisely.**** g_e sees only the totally split primes; each g_h sees only the primes with Frobenius h . This is a ****Chebotarev resolution of the screw kernel**** — an arithmetic decomposition with no counterpart for ζ . It is ****not**** a claim that g_h is itself a screw function: a signed combination of conditionally positive definite functions need not be conditionally positive definite, and we make no such claim. The correct name is ***Frobenius-resolved kernel component***.

§8.2 Theorem T11 — the Channel Perturbation Theorem [PROVEN]

Suzuki's finite-interval theorem (I5) is proved for ζ . It does not follow, and cannot simply be assumed, that for each Dirichlet character the localised form is closable, lower semibounded, has a form core and a compact resolvent. **We prove it, by a bounded-form perturbation whose norm we compute.**

Lemma T11a (the singularities cancel) [PROVEN]. For $x \neq 0$, $\omega_c(x) = e^{-2c|x|}/(1 - e^{-2|x|}) \sim 1/(2|x|)$, the **same** leading singularity for every c . In the difference the singularity cancels exactly:

$$\omega_{\{c\chi\}}(x) - \omega_{\{1/4\}}(x) = [e^{-2c|x|} - e^{-|x|/2}] / (1 - e^{-2|x|}) \rightarrow **1/4 - c\chi** \text{ as } x \rightarrow 0,$$

so the difference extends to a bounded continuous function on $\mathbb{R}$. Moreover $\psi := \Phi_{\{c\chi\}} - \Phi_{\{1/4\}}$ (the antiderivative pair of §3) has a **finite** one-sided derivative $\psi'(0+) = 1/2(\psi_{\{dg\}}(c\chi) - \psi_{\{dg\}}(1/4))$, because $\sum_k [1/(k+c) - 1/(k+1/4)]$ converges absolutely — whereas each Φ separately has a logarithmically divergent derivative at 0. Hence $T_{\{c\chi\}} - T_{\{1/4\}} = (\omega_{\{c\chi\}} - \omega_{\{1/4\}}) \cdot dx + 2\psi'(0+) \cdot \delta_0$.

Lemma T11b (the difference kernel has a finite second derivative measure) [PROVEN]. Put $h_\chi := g_\chi - g_\zeta$. Then on any interval $[-2a, 2a]$,

$$h_\chi'' = v_{\{\chi, a\}} = **(-\log q_\chi) \cdot \delta_0** + \sum_{\{n \leq e^{2a}\}} (\chi(n)-1)\Lambda(n)n^{-1/2}(\delta_{\{\log n\}} + \delta_{\{-\log n\}}) + [\omega_{\{c\chi\}} - \omega_{\{1/4\}} + e^{|x|/2} + e^{-|x|/2}] \cdot dx,$$

a **finite signed Radon measure**: finitely many prime atoms (only $n \leq e^{2a}$ contribute on this interval), one point mass, and a bounded density. The δ_0 coefficient is exactly $-\log q_\chi$, because $2(\beta_\chi - \beta_\zeta) + 2\psi'(0+) = -\log q_\chi$ — the digamma terms cancel and only the conductor survives. [VERIFIED to 10^{-8} : $-1.09861 = -\log 3$, $-2.39790 = -\log 11$, $-3.49651 = -\log 33$.]

Theorem T11 (Channel Perturbation) [PROVEN]. Let $u \in C_c^\infty(-a, a)$ and $U := u * \tilde{u}$. Since Du has mean zero and $(Du) * (Du)^\sim = -U''$, the localised Weil forms differ by

$$q_\chi[u] - q_\zeta[u] = -\langle h_\chi'', U \rangle = -\langle v_{\{\chi, a\}}, U \rangle, \text{ and } |U(t)| = |\langle \tau_t u, u \rangle| \leq \|u\|^2,$$

hence **** $|q_\chi[u] - q_\zeta[u]| \leq \|v_{\{\chi, a\}}\|_{TV} \cdot \|u\|^2$ **** — an **** L^2 -bounded**** symmetric form perturbation.

The perturbation as a bounded operator (the step written out). Define the sesquilinear form

Let $E_a : L^2(-a, a) \rightarrow L^2(\mathbb{R})$ be extension by zero, $P_a : L^2(\mathbb{R}) \rightarrow L^2(-a, a)$ restriction, τ_t translation, and set $T_t := P_a \tau_t E_a$, a contraction on $L^2(-a, a)$ with $\|T_t\| \leq 1$ for every t (this is the point at which the interval must be handled: τ_t alone does not map $L^2(-a, a)$ to itself). Define

$$b_{\{\chi, a\}}[u, v] := - \int_{\{-2a, 2a\}} \langle T_t u, v \rangle \cdot dv_{\{\chi, a\}}(t), \quad u, v \in L^2(-a, a),$$

Note $\langle T_t u, v \rangle = (u * \tilde{v})(-t)$ up to conjugation, so this is exactly the pairing that appears in the form difference.

Since $|\langle T_t u, v \rangle| \leq \|T_t\| \cdot \|u\|_2 \|v\|_2 \leq \|u\|_2 \|v\|_2$ for every t , we get $|b_{\{\chi, a\}}[u, v]| \leq \|v_{\{\chi, a\}}\|_{TV} \cdot \|u\|_2 \|v\|_2$, so b is a

bounded sesquilinear form; it is symmetric because v is real and even and $T_t^* = T_{-t}$. By the Riesz representation theorem there is a **unique bounded self-adjoint** $B_{\{\chi, a\}}$ on $L^2(-a, a)$ with $b_{\{\chi, a\}}[u, v] = \langle B_{\{\chi, a\}} u, v \rangle$ and $\|B_{\{\chi, a\}}\| \leq \|v_{\{\chi, a\}}\|_{TV}$. Taking $u = v$ recovers the quadratic estimate above.

Compact resolvent, written out. For z real and sufficiently negative, $\|B_{\{\chi, a\}}(A_a^\zeta - z)^{-1}\| < 1$, and

$$(A_a^\chi - z)^{-1} = (A_a^\zeta - z)^{-1} \cdot [I + B_{\{\chi, a\}}(A_a^\zeta - z)^{-1}]^{-1}.$$

The first factor is compact (Suzuki, I5) and the second is bounded, so the product is compact. Hence A_a^χ has compact resolvent, discrete spectrum and finite multiplicities, and $D(A_a^\chi) = D(A_a^\zeta)$.

Consequences (KLMN + bounded perturbation), all unconditional: (i) q_χ is closed and lower semibounded on the **same form domain** as q_ζ , with lower bound shifted by at most $\|v_{\{\chi,a\}}\|_{TV}$; (ii) the associated self-adjoint operator is $A_a^\chi = A_a^\zeta + B_{\{\chi,a\}}$ with B symmetric and $\|B\| \leq \|v_{\{\chi,a\}}\|_{TV}$; (iii) a bounded perturbation preserves compact resolvent, so A_a^χ has **discrete spectrum with finite multiplicities**; (iv) by Weyl's inequality

$$**|\lambda_n(A_a^\chi) - \lambda_n(A_a^\zeta)| \leq \|v_{\{\chi,a\}}\|_{TV}** \quad \text{for every } n.$$

Table 8.1. The perturbation norm $\|v_{\{\chi,a\}}\|_{TV}$ (from the closed form of Lemma T11b; prime cutoff $n \leq \lfloor e^{2a} \rfloor$). It is dominated by the prime atoms and grows like $O(e^a)$, as it must: the prime part of g at $|t| \leq 2a$ carries every prime power up to e^{2a} . **These are VERIFIED floating-point values of a PROVEN finite bound**, not ball-arithmetic certificates.

a	χ^{-3}	χ^{-11}	χ^{33}	$\lfloor e^{2a} \rfloor$
1	14.877	14.970	19.988	7
2	53.748	52.040	57.087	54
3	154.351	154.763	160.643	403
4	432.527	431.399	436.364	2980
6	3221.744	3221.248	3226.716	162754
8	23841.094	23844.519	23847.045	8886110

What this closes. The channelwise operator theorem was, in the previous version, an assumed hypothesis (O5). It is now a **theorem**, obtained without redoing any of Suzuki's analysis: the only input is his ζ -theorem plus the observation that the archimedean singularity is *the same* in every channel and therefore cancels in the difference. **This is the step that extends Suzuki's finite-interval operator theory from ζ to the real primitive Dirichlet characters, and hence to the Dedekind zeta functions of multiquadratic abelian fields** — those whose entire character group consists of real characters, which is exactly the case for $K = \mathbb{Q}(\sqrt{-3}, \sqrt{-11})$. **Scope**, stated precisely: a general finite abelian extension has complex characters occurring in conjugate pairs $\chi, \bar{\chi}$ for those the kernel g_χ need not be real and even, and a realified conjugate-pair block must be built before the same perturbation argument applies. That construction is straightforward in principle but is **not carried out here**, so T11 as proved covers the real primitive characters and the multiquadratic fields they generate.

§8.3 Theorem T12 — the Galois-graded operator [DERIVED, now unconditional]

With T11 in hand the graded operator is unconditional. Assemble the Frobenius-resolved components into $\mathcal{M}((x,h),(y,h')) := g_{\{h h'^{-1}\}}(x-y)$ on $L^2(-a,a) \otimes \mathbb{C}[G]$, and set $\mathfrak{A}_a := D^* \mathfrak{G}_a D$ (mean-zero compression, $D = i \, d/dx$ Dirichlet).

Theorem T12 [DERIVED]. The finite Fourier transform on G block-diagonalises $\mathfrak{A}_a$ into $\oplus_{\chi \in \hat{G}} A_a^\chi$. By T11 each block is self-adjoint, lower semibounded and discrete; therefore $\mathfrak{A}_a$ is **self-adjoint, lower semibounded, and has discrete spectrum**. It commutes with the regular representation of G — a symmetry the ζ operator does not possess — and its spectrum is **graded by $\hat{G}$** , the grading being the Frobenius grading of the arithmetic.

Suzuki's A_a is the trivial-character block. $\mathfrak{A}_a$ is the **canonical finite direct-sum operator containing all four character blocks** — the smallest such assembly in the obvious sense that it has exactly one block per element of $\hat{G}$ — and it is obtained at the cost of one perturbation lemma. No universal property is claimed.

§8.4 What the grading detects — and what we actually observed

Under GRH(ζ_K) and a channelwise spectral limit (T11 now supplies the operator theorem itself), the limiting multiplicity of γ in $\text{Spec}(\mathfrak{A}_\infty)$ would be $\sum_{\chi \in \hat{G}} m_\chi(\gamma)$, the total analytic multiplicity across the four L-functions. A multiplicity ≥ 2 therefore signals **either** a zero shared by two channels **or** a repeated zero within one channel; separating the two requires the additional hypothesis that each channel's zeros are simple.

What we observed is a statement about zeros, not about an operator. Comparing the sign-change zeros detected below $T = 40$:

Table 8.2. Detected sign-change zeros below $T = 40$, and the closest approach between distinct channels. This is a **numerical observation about a detected list**, not a certification: no Turing-type zero count was run, so undetected or even-multiplicity zeros are not excluded.

channel	detected zeros, $T < 40$	lowest certified γ
ζ	6	14.134725
χ^{-3}	13	8.039737
χ^{-11}	21	2.477244
χ^{33}	28	2.996951
closest approach across distinct channels	ζ at 25.010858 vs χ_{33} at 25.011945	$\Delta = 1.087 \times 10^{-3}$

No coincidence occurs among the detected zeros below $T = 40$, and the closest approach is 1.1×10^{-3} — a near-degeneracy in the very first range one looks at. We do **not** claim this certifies simplicity or the Grand Simplicity Hypothesis for this family; it is an observation, and it is recorded because a **limiting** graded operator — if the channelwise spectral limit exists — would see it as a splitting between two $\hat{G}$ -blocks (the finite- a graded operator itself exists unconditionally by T12).

§8.5 Conditional outlook — POME, and what the pre-gate does and does not show

The kink measure (T4) makes the corpus’s dynamical target explicit: a flow whose primitive orbits carry a Galois holonomy and reproduce μ_{Λ}^{χ} for every χ (POME-K). A necessary condition is the orbit count $N_{IT}(L) \sim \pi(e^L) \sim e^L/L$. By Parry–Pollicott that count is **automatic** for any weak-mixing Axiom-A flow of entropy 1, so **the counting condition does not by itself decide POME** — an earlier internal claim to the contrary is retracted. **What the pre-gate does establish [DERIVED]**. The corpus’s map is $f(z) = i^z = \exp((i\pi/2)z)$, whose fixed points are exactly $z_k = -(2/i\pi) \cdot W_k(-i\pi/2)$ over the branches $k \in \mathbb{Z}$ of the Lambert function. Since $|W_k(w)| \sim 2\pi|k|$, we get $|z_k| \sim 4|k|$ and hence

$N_{fix}(R) := \#\{\text{fixed points with } |z| \leq R\} \sim R/2$ (asymptotically; the argument principle gives 3, 5, 10, 20 at $R = 5, 10, 20, 40$).

So f has infinitely many fixed points, and hence infinitely many periodic points in total. (Whether it has infinitely many points of **every exact period** is **not** established here: the Lambert-W computation solves $f(z) = z$ only, and the period- n count would require analysing $f^n(z) = z$. The external-address heuristic suggests it, but we do not claim it.) Consequently, **if the roof function is bounded on the invariant set**, infinitely many orbits have length below some finite L , contradicting $N_{IT}(L) < \infty$. POME therefore requires **either** a roof unbounded on the invariant set **or** a pruning selecting finitely many orbits per period — and, to avoid circularity, the pruning must be prime-blind.

Status, stated exactly. ‘The unpruned map has infinitely many periodic points’: **DERIVED** (Lambert-W branches). ‘Its topological entropy on the Julia set is infinite’ and ‘no prime-orbit asymptotic can hold’: **OPEN** — the radius R is not an orbit length, and entropy is not determined by a fixed-point count. An earlier version asserted these as **PROVEN**; that is **retracted**. What remains is a sharp open question: **is there a canonical, prime-blind pruning of the i -tetration dynamics, with an unbounded roof, whose primitive orbit lengths are exactly $\{\log p\}$?**

§8.6 A remark on operator routes

A Hilbert-space Gram realisation of the screw kernel, $K_g(t, u) = \langle b(t), b(u) \rangle$, exists **if and only if** $K_g \ni 0$ — which, globally, is RH. So **asserting the existence of such a realisation** is asserting the positivity, not explaining it. This does **not** say that operator-theoretic routes are futile: Suzuki’s A_a exists as a self-adjoint, lower-semibounded operator **without** assuming RH (its spectrum may a priori contain negative values), and proving its positivity by independently established operator structure would be a genuine proof. An earlier version claimed that no operator built from g can be a route and that only a dynamical construction remains; **both claims are withdrawn** as overreach.

§9. Conclusion

Three things are established. The screw kernel of an abelian field decomposes over the character group and detects Frobenius (§2). Its second distributional derivative is a decorated prime-orbit measure, which turns the dynamical programme into an explicit and falsifiable target (§3). And the Weil functional of K can be **certified** —

positive on the canonical grid without GRH, and positive definite on an 11-dimensional space of test functions — by ball arithmetic with closed-form tails (§4–§5).

The fourth is the one to take away, and it is a statement about computation rather than about zeta functions. Because the zero side is a sum of squares, a **single** zero gives a rigorous lower bound; the budget of a finite Weil audit is therefore set by the **nearest certified usable zero among the channels**. K supplies a certified low-lying zero at 2.477 where the smallest certified sign-change zero of ζ found in the declared scan is 14.135 (firstness is not claimed anywhere; see §6.1), and that single fact moves the certified prime budget of the worst grid point from 10^{32} to 10^6 . **A character decomposition containing a low-zero channel is a numerical preconditioner**. Whether enlarging a character family reliably supplies such a channel — the family statement — is open, and the naive conductor heuristic is false. What is established is the mechanism and the case.

One further thing is established. §8.1 resolves the screw kernel by **Frobenius class** rather than by character: g_h sees exactly the primes with Frobenius h , so g_e sees exactly the totally split primes. This is a Chebotarev decomposition of the kernel with no counterpart for ζ , and it is unconditional.

§8.1–§8.3 are established results, not an outlook. §8.1 resolves the kernel by Frobenius class (T10, PROVEN). §8.2 **transfers Suzuki’s finite-interval operator theorem from ζ to every real primitive Dirichlet channel** (T11, PROVEN): the archimedean singularity $1/(2|x|)$ is the same in every channel and cancels in the difference, so h_χ'' is a finite signed measure with δ_0 -mass exactly $-\log q_\chi$, the localised forms differ by an L^2 -bounded symmetric form, and KLMN plus the resolvent identity give a self-adjoint A_a^χ with the same form domain, compact resolvent, discrete spectrum, and $|\lambda_n(A_a^\chi) - \lambda_n(A_a^\zeta)| \leq \|v_{\{\chi,a\}}\|_{TV}$. §8.3 assembles the channels into a **Galois-graded self-adjoint operator $\mathfrak{A}_a$** (T12, DERIVED, unconditional given T11), G -equivariant and $\hat{G}$ -graded. **The channelwise operator hypothesis that earlier versions carried as OPEN is closed.**

§8.4–§8.6 remain observational or conditional, and are labelled so. The closest approach between the detected zeros of distinct channels below $T = 40$ is 1.1×10^{-3} — an observation about a detected list, not a certification, since no Turing-type count was run. The dynamical route (POME-K) is constrained but not decided: the unpruned map provably has infinitely many periodic points (Lambert-W branches), while the entropy and orbit-count conclusions remain open. And the Gram remark of §8.6 stands as a caution, not a no-go.

What is not established is stated as such. The operator frontier is open, its two tracks are separated, and the single missing statement of the Herglotz route — the Weyl–truncation bridge — is written down. The next paper in this line will be written when one of those changes status, and not before.

Acknowledgements & Code Availability

Reproducibility record

item	value
script	zs_m48_verify_v2_7.py
SHA-256	366e9c0e247daa65ff87b1e4200e9d1b1f3ec37e54275081848065c172a9d3c8
expected checks	**380** (hard-coded; 360 with --fast, which is a smoke test, not the certificate). v2.6 advertised 378 but had added two P3 checks ($a = 3, 6$) without updating the guard; corrected here. A full run must confirm 380 — and if it does not, the hard-coded guard **fails loudly** , which is its purpose.
Python	3.12.3
numpy / scipy	2.4.4 / 1.17.1
mpmath / python-flint (arb)	1.3.0 / 0.9.0
still required before public release	the stdout and JSON ledger of a full 380/380 run, and its wall-clock time

The companion script zs_m48_verify_v2_7.py is a single self-contained fail-closed suite. Every theorem-tier certificate (the enclosures of §4, the real-zero certification of §4.3, the zero brackets, the Cholesky certificate of §5, the budget inversions of §6) is formed and decided inside arb ball arithmetic; floats appear only in the JSON

report. The Herglotz and pilot-assembly blocks are labelled **diagnostics** and are not certificates. The script prints its OPEN registry, refuses to report a check count different from the advertised one, and exits non-zero on the first failure. This work used AI tools for verification, adversarial audit and drafting; the author is responsible for all content.

Appendix A. Specialisation of the zero-free formula

[2, Thm 4.1] with $r = 1$, $\lambda = \frac{1}{2}$, $\mu = a\chi/2$, $Q = (q\chi/\pi)^{1/2}$, $m_F = 0$ (1 for ζ), $c_F(n) = \chi(n)\Lambda(n)$ gives $\lambda/2 + \mu = c\chi$, $\lambda^2 = \frac{1}{4}$, $e^{\{-t(\frac{1}{2}+\mu/\lambda)\}} = e^{\{-2c\chi t\}}$, $e^{\{-t/\lambda\}} = e^{\{-2t\}}$, and $\log Q + \lambda\psi(c\chi) = -\beta\chi$ — which is Theorem T2.

Appendix B. Closed forms

$F(x) = C e^{\{-ax^2/2\}}(\cos tx + \kappa)$, $C = \frac{1}{2}\sqrt{(\pi/2a)}$, $\kappa = e^{\{-t^2/2a\}}$; $\hat{v}(z) = \frac{1}{2}\sqrt{(\pi/a)}[e^{\{-(z-t)^2/4a\}} + e^{\{-(z+t)^2/4a\}}]$. Prime part, exact: $\int(|x|-L)\cdot F''dx = 2F(L)$. Linear part: $\int\beta|x|F''dx = 2\beta F(0)$. Gamma-factor part: $-\frac{1}{2}\sum_k J(2\mu_k)/\mu_k^2$ with $J(\beta) = -\beta F(0) + \beta^2 G(\beta)$, $G(\beta) = C[\Re H(\beta-it) + \kappa H(\beta)]$, $H(w) = \sqrt{(\pi/2a)}\cdot\text{erfcx}(w/\sqrt{(2a)})$ (Faddeeva; no cancellation). Pole part (ζ only): $2\int e^{\{x/2\}}F dx$, a Gaussian integral.

Appendix C. The subspace Gram matrix

Multiplying two Gaussian-cosine transforms and using $e^{\{-(z-p)^2/4a\}}e^{\{-(z-q)^2/4a\}} = e^{\{-(p-q)^2/8a\}}e^{\{-(z-(p+q)/2)^2/2a\}}$, then inverting, gives the closed form of §5. Hence every Gram entry is a combination of two evaluations of $\Phi(a, \omega) = -\int\chi(x)[e^{\{-ax^2/2\}}\cos(\omega x)]''dx$, and the enclosure machinery of §4 applies verbatim.

Appendix D. Verification ledger and Supplement

The full ledger, the 48-entry table, the per-channel verdicts, the zero brackets, the anti-numerology Monte Carlo (0/200 successes per null model in each of three channels; 95% upper bound $3/200 = 1.5\%$ by the rule of three), the pilot assembly outputs, and the complete **Correction History** of drafts v1.0–v1.9 — including every retraction — are in the Supplement. Nothing in the body depends on the Supplement; the Supplement exists so that the corrections are on the record.

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Version History

v2.7 (July 2026): **release candidate**. Corrigendum to v2.6, with no content removed. Fixes: the fail-closed guard is corrected from 378 to **380** (v2.6 added two P3 checks at $a = 3, 6$ without updating it — a full run would have

failed); T11's scope is stated precisely (real primitive Dirichlet characters, hence **multiquadratic** abelian Dedekind zetas; complex conjugate-pair channels require a separate realified block, not carried out here); the translation used in the sesquilinear form is defined as the contraction $T_t = P_a \tau_t E_a$ (zero extension then restriction), since τ_t alone does not preserve $L^2(-a,a)$; the claim of infinitely many periodic points **of every exact period** is retracted (only the total set is shown infinite); 'Four results' $\rightarrow$ 'Five'; the duplicate Table 8.1 is renumbered 8.2; §7's opening sentence no longer says everything in it is OPEN; §8.4 refers to the **limiting** graded operator; 'no new operator analysis is required' $\rightarrow$ 'no new compactness proof is required for each channel'; and the remaining 'lowest zero' wording is replaced by 'certified low-lying zero / smallest certified sign-change zero found in the declared scan'. **Status: release candidate, not yet a public release** — the stdout, JSON ledger and wall-clock of a full 380/380 run are still required.

v2.6 (July 2026): corrigendum to v2.5, with no content removed. Fixes: the Conclusion now states §8.1–§8.3 as established (it had still called them a conditional outlook, contradicting T11); the OPEN registry no longer lists the channelwise operator hypothesis (T11 closes it); T11 gains the explicit sesquilinear-form, Riesz-representation and resolvent-identity paragraphs; the perturbation norms are relabelled VERIFIED rather than arb-certified (the theorem is PROVEN, only its numerical bound is floating-point); the prime cutoff in Table 8.1 is corrected to $n \leq \lfloor 2a \rfloor$, changing the $a = 1$ entries from 15.86/15.95 to **14.877/14.970**; $a = 3$ and $a = 6$ are added to the verification loop so every table row has provenance; the T8 cutoff in the contributions table is corrected to $P = 4 \times 10^6$; β_χ is defined by the gamma-factor expression (with the L'/L identification used only where $L(\frac{1}{2}, \chi) \neq 0$ is certified); 'smallest self-adjoint operator' is replaced by 'canonical finite direct-sum operator'; and the remaining first/lowest-zero wording is unified on γ_{cert} .

v2.5 (July 2026): draft. **Proves the Channel Perturbation Theorem (T11)**: the archimedean singularity $1/(2|x|)$ is the same in every channel and cancels in $h_\chi = g_\chi - g_\zeta$, so h_χ'' is a finite signed measure on $[-2a, 2a]$ with δ_0 -mass exactly $-\log q_\chi$; the localised forms therefore differ by an L^2 -bounded symmetric form, and Suzuki's ζ operator theorem transfers to every real primitive Dirichlet channel by KLMN, with $|\lambda_n(A_a^\chi) - \lambda_n(A_a^\zeta)| \leq \|\chi_a\|_{\text{TV}}$. This closes the former hypothesis O5 and makes the Galois-graded operator (T12) unconditional. Also: **retracts** 'the T9 remainder is independent of a ' (only the $1/d^2$ coefficient is; the next is $-(1+aL^2)/2d^3$, now given); unifies the certificate cutoff at an explicit $P = 4 \times 10^6$ plus analytic tail; removes the environment override of the check-count guard; and separates theorem-tier from diagnostic blocks in the ledger.

v2.4 (July 2026): draft. **Closes Theorem T9**: the $1/d^2$ remainder is now derived (from the complementary-error-function factor of the tail) rather than measured, is independent of a , and reduces the error to 2×10^{-4} nats where $\log N^* \approx 73\text{--}97$. **Narrows T10** to unramified prime powers and renames it a Frobenius-*resolved kernel component* (a signed combination of cpd functions need not be cpd, and no screw property is claimed). **Demotes the Galois-graded operator, the shared-zero statement and the POME pre-gate to a labelled conditional outlook** (§8.2–8.4), with three explicit retractions: the channelwise operator theorem is **not** inherited from Suzuki; the multiplicity formula is $\sum \chi m_\chi(\gamma)$ and does not by itself separate shared from repeated zeros; and neither infinite entropy nor the impossibility of a prime-orbit asymptotic was proved. **Withdraws** the claim that no operator built from g can be a route to RH, and the claim that only a dynamical construction remains.

v2.3 (July 2026): draft. Corrects Theorem T9 (the target prefactor carries $(1 + e^{\{-\gamma t/a\}})^2$ — a factor 4 at $t = 0$ — and the subleading term is $(1/d)\log(A/B)$, not $(a/d)\log(\cdot)$; the corrected law is accurate to 0.005 nats where $\log N^* = 73$, with residual $\approx 4.5a/d^2$, and its regime $d^2/a \rightarrow \infty$ is stated). Replaces all 'first zero' language by 'certified zero'. Adds §8 (the Galois-twisted screw operator: T10 Frobenius-class kernel, T11 the operator, T12 simplicity detection, T13 the no-go for g -built operators) and §7.3.1 (the POME pre-gate, executed: the unpruned i -tetration dynamics has infinitely many fixed points, $N(R) = R/2$, hence infinite entropy and no prime-orbit asymptotic — and a retraction of the earlier claim that the counting condition alone decides POME).

v2.2 (July 2026): draft. Adds the certification threshold law (Theorem T9); replaces the 'first zero' language by 'a certified low-lying zero' (a sign change certifies an odd-multiplicity zero in the bracket, not the absence of lower

ones — and T7/T8 do not need firstness); makes the budget inversion fail-closed with two-sided postconditions; decides the gain entirely in arb; states the prime cutoff of the certificate exactly (explicit summation to 4×10^6 plus the closed-form tail). Compact rewrite. Versions v1.0–v1.9 were internal audit drafts and are superseded; their corrections — including the retraction of a false positivity certificate for A_a , of an over-read zero-side lower bound, and of an incorrect Nevanlinna convergence criterion — are preserved in the Correction History Supplement rather than in the body. The new result of this release is §6, the Decoration Principle.